

Research on MEMS fast steering mirror technology requirements for space optical communication system

Chao Wu*, Bin Ren, Huijian Liang, Jingjing Xue, Xiang Chen, Haike Li, Xingyu Huo
Xi'an Institute of Space Radio Technology, Xi'an, Shaanxi 710000, China

ABSTRACT

Miniaturization and low-cost laser communication terminal is the development direction of space payload. MEMS fast steering mirrors (FSM) has the characteristics of small size, low power consumption, large deflection angle, high closed-loop control bandwidth and low cost. It is a potential fast steering mirror to realize the miniaturized laser communication terminal beam high-precision scanning and pointing, and is also one of the development directions of low SWaP-C laser communication terminal technology. For space laser communication terminal pointing acquisition and tracking (PAT) system, the optical and mechanical characteristics of MEMS fast steering mirror used in beam deflection mechanism are analyzed in combination with theory and engineering application. According to the quasi-static working mode of MEMS fast steering mirror for fine pointing assembly (FPA) and point ahead angle (PAA) mechanism and the resonant working mode for nutation coherent tracking mechanism, the static and dynamic plane types, working bandwidth and scanning angle, closed-loop control accuracy, volume, weight, power consumption and reliability requirements of MEMS fast steering mirror are analyzed in detail. The technical parameters of MEMS FSM for space laser communication are presented, which can provide reference for its application in space laser communication system.

Keywords: MEMS fast steering mirror, pointing acquisition and tracking, optical and mechanical characteristics

1. INTRODUCTION

In recent years, under the traction of large-scale low-orbit satellite constellation networking applications, countries are competing to develop low SWaP-C (Size, Weight and Power) laser communication terminals suitable for commercial applications. Typical of the Starlink constellation in the United States, LeoSat constellation and Telesat constellation in Canada, such laser terminals usually weigh 10kg, the communication rate is 10Mbps to 10Gbps, and the power consumption is about tens of watts[1~3].

Pointing Acquisition and Tracking (PAT) system is an important part of laser communication terminal and the core of realizing stable chain construction of laser communication terminal. The Fast Steering Mirror (FSM) has the advantages of high deflection frequency, high pointing accuracy, and high resolution. It is used as a precision photoelectric component to control the beam direction between the receiving and sending ends of the optical path in the space laser communication terminal. It is a central component of the PAT system.

The traditional FSM solutions are piezoelectric ceramic FSM and voice coil motor driven FSM. The piezoelectric ceramic (PZT) FSM, which uses the inverse piezoelectric effect of the material to provide the driving force, has the advantages of small size, large driving force, high resolution and fast response, but its shortcomings are poor impact resistance, piezoelectric ceramics have nonlinear characteristics such as dielectric and hysteresis, often need to enlarge the structure to expand the displacement, and large volume, weight and power consumption. The electromagnetic FSM driven by voice coil motor, which provides driving force by the current conductor subjected to Lorentz force in the magnetic field, has the advantages of large travel, low driving voltage and high bearing capacity, but its disadvantage is that it is easy to be affected by the interference of ambient temperature and magnetic field, and the response frequency is often lower than that of the piezoelectric ceramic FSM[4~6].

FSM prepared based on Micro Electro Mechanical System (MEMS) technology have the advantages of extremely low mass and volume, low power consumption and close integration with electronic equipment, which can reduce the volume and mass of spacecraft without sacrificing the function. However, the optomechanical feasibility of its application in the field of space laser communication has not been fully demonstrated. In this paper, the working principle and driving mode of MEMS FSM are introduced. Based on the requirements of MEMS FSM used in space laser communication terminal PAT, the optical and electromechanical requirements are comprehensively analyzed. The feasibility of MEMS FSM applied in satellite laser communication is demonstrated through comprehensive analysis.

2. MEMS FSM WORKING MODE

MEMS FSM has a variety of types such as electric thermal drive, piezoelectric drive, electrostatic drive and electromagnetic drive. In view of the advantages of simple structure, low power consumption, fast response speed, high frequency, and good process compatibility, electrostatic drive is mostly used, and mostly implemented by flat plate capacitor. The technology is more mature and widely used, such as the unique universal-jointless two-axis FSM of Mirrorcle Technologies in the United States. Electrostatically driven MEMS scanning mirror is generally composed of two parts: driver and mirror. In general, the driver determines the electromechanical properties of the scanning mirror, while the mirror determines the optical properties of the scanning mirror[7~9].

MEMS FSM can be divided into resonant working mode and quasi-static working mode. Resonant MEMS FSM works in mechanical resonance state. It has large scanning angle, low drive power consumption and low scanning current. The image beam scanning and laser imaging are its main application markets. Quasi-static MEMS FSM works in a non-resonant state, and the FSM can be paused at any deflection angle within the scanning range. It is mainly used in laser pointing and laser vectorized graphics scanning, and the scanning angle range is relatively small.

For PAT system, it generally includes coarse pointing assembly (CPA), FPA mechanism, PAA mechanism, acquisition and tracking photoelectric detector, communication photoelectric detector, etc. With the miniaturization design requirements, the communication and tracking integration technology based on nutator coherent tracking is gradually used, so nutation FSM is also needed. FPA is used to suppress the vibration of the platform, track the target with high precision, and establish and maintain the communication link. PAA is used for advanced targeting of the tracking mechanism due to orbital motion. According to the use requirements of FPA and PAA, FPA and PAA can choose the same FSM, and they need to work in quasi-static mode. Nutator needs to work in quasi-static mode for optical coaxiality adjustment and resonance mode for nutation.

FPA is the first level executive mechanism to realize the beacon light tracking function. The FPA, which has large deflection range, high execution precision and large closed-loop bandwidth, is used as a first-level actuator to suppress the vibration of the platform. In the capture stage, FPA works with the capture sensor to complete scanning and rough calibration of errors. In the tracking stage, FPA and tracking sensor compensate the vibration of the satellite platform to achieve high-precision laser beam pointing. According to the uncertain region and the capture and tracking system, the field of view of the FPA pre-stage optical telescope is generally 4mrad, and the magnification is 10 times, so the FPA field of view needs to be more than 400 μ rad.

PAA is an executive mechanism of terminal advance aiming, which is used to compensate the advance angle caused by the light wave propagation direction of both communication parties due to the free space transmission delay. According to the different inter-satellite links, the angle range of advanced aiming can be from 30 μ rad to 100 μ rad. In the scheme without beacon tracking, the angle of this order is large relative to the divergence angle of communication beam, so PAA without beacon light tracking must have higher pointing accuracy than PAA with beacon light tracking.

Nutation coherent tracking is a high resolution solution to the two-dimensional detection position and off-target information from the one-dimensional communication detector. It uses a two-dimensional mechanism and a one-dimensional detector to achieve high performance two-dimensional detection. The signal intensity comes from the coherence intensity between the local oscillator and the signal. Signal or local oscillator focus on the detector photosensitive surface above a certain radius of high-speed and high-precision circular motion, only when the received signal light focus is located in the center of the local oscillator focus or local oscillator focus is located in the center of the signal light focus track, the output voltage signal is stable, otherwise the output voltage is fluctuating. Through the fluctuation of synchronous coherence efficiency and the movement information of local focus, the error of the signal visual axis is extracted.

3. MEMS FSM TECHNICAL REQUIREMENTS

3.1 Size of scanning mirror

The mirror size of the MEMS FSM needs to be determined according to the needs of the beam scanning system. In the specific design, according to the requirements of the scanning system resolution, combined with the product of the mirror size and the maximum scanning angle, the appropriate processing mirror size can be determined.

Specifically, the angular radius of the reflected spot can be described by the classical airy spot, which is $1.22\lambda/D$, that is, proportional to the wavelength and inversely proportional to the mirror size. When the size of the reflected spot is determined, the resolution of the MEMS mirror can be obtained by dividing the maximum scanning angle by the single

point size of the spot, that is, the approximate $D/(1.22\lambda)$. In practice, the diameter of the optical mirror should be greater than

$$d > \sqrt{2} \frac{D}{\beta} \quad (1)$$

In the above formula, D is the receiving optical aperture, β is the magnification of the telescopic unit, and $\sqrt{2}$ is the correction factor of placement at a 45 degree angle. For the requirements of mirror thickness, it is mainly based on the optical material, to ensure a certain surface quality in high-speed servo, usually more than 0.25 diameter. Combined with the engineering requirements of space laser communication optical system, the mirror size of MEMS FSM is greater than or equal to 5mm.

3.2 Scanning frequency

According to the TerraSAR vibration spectrum of the satellite platform, the vibration energy of the satellite platform is mainly concentrated below 100Hz. Under vibration conditions, high-precision beam control is still required to maintain the stability of the link, so the design bandwidth of the MEMS FSM of the fine tracking mechanism in the working quasi-static mode requires more than 2kHz to suppress the vibration brought by the satellite platform. In the entire bandwidth range, closed-loop execution error is required to be no more than 20μrad within the MEMS central region range ($\pm 3\text{mrad}$) when tracking the state.

The resonant operating mode is generally used in the coherent tracking sensor of laser communication to carry out high-frequency nutation of the transmitted signal light, and the nutation frequency is the resonant frequency. In theory, the higher the nutation frequency, the better, but in practice the nutation frequency is limited by the response capability of the nutation device and the processing speed of the subsequent circuit. Small nutation frequency will affect the ability of laser subsystem to suppress the vibration of satellite platform. Taking the TerraSAR vibration spectrum as an example, when the nutation frequency reaches more than 5kHz, the normalized mutual relation number is greater than 0.99. Combined with engineering experience, the position detection bandwidth of the nutation coherent tracking sensor is close to that of non-coherent tracking.

For mechanical vibration loads, the most important factor is the coupling of the frequency ω of the applied vibration with the natural frequency ω_0 of the MEMS structure, so that the applied mechanical force F_0 is amplified as [10]

$$F = F_0 \left[\left(1 - \frac{\omega^2}{\omega_0^2} \right)^2 + \frac{\omega^2}{Q^2 \omega_0^2} \right]^{-1/2} \quad (2)$$

In equation (2), Q is the quality factor under a specific mode, and its value less than 1 is conducive to eliminating or reducing the force amplification caused by the coupling effect. Therefore, the resonance frequency of the MEMS fast mirror should be higher than that of the transport and transmission, and the higher the resonance frequency, the better the seismic performance.

3.3 Scanning angle

In the PAT system, CPA mechanism conducts a large range of orbital motion pointing, and the way of real-time unloading is used by CPA to make up for the shortcomings of small range of FPA. The angle range of the FPA should cover the tracking field of view. The coarse tracking unit ensures that the light spot enters the fine tracking field of view reliably, which requires that the control angle range of the fine tracking servo unit must be larger than the fine tracking field of view. Generally, FPA is installed at 45°, then the quantitative expression of the angle execution range of FPA is

$$\alpha_{FSM} = \frac{\theta_{FOV} \cdot \beta}{2} \quad (3)$$

In the above equation, β is the magnification of the telescopic unit, θ_{FOV} is the fine tracking field of view. It can be seen that the larger the magnification of the cassette system, the wider the angle servo range of the galvanometer is required. If the field of view of the fine tracking unit is 300μrad and the magnification of the optical telescopic system is 10 times, the minimum execution range of the fine tracking galvanometer should be greater than 1.5mrad.

For nutation mechanism, the deflection range mainly depends on the adjustment range of the coaxiality of the optical axis and the size of the nutation half angle. The selection of nutation half angle mainly considers two factors: the coherence efficiency loss as small as possible and the angle error signal extraction sensitivity as high as possible. In the

actual design, the nutation half angle can also meet the angle error extraction under the condition of reducing the coherence efficiency loss.

After the space laser communication terminal is launched into orbit, due to the vibration of the rocket during launch and the thermal deformation of the space, the transeiving coaxiality of the optical system will change compared with the ground, and the general deviation is less than 20 μ rad. According to the engineering needs, considering the coaxiality calibration and nutation half angle, the mechanical deflection range of nutation mechanism can meet the requirements when it reaches 100 μ rad.

3.4 Execution accuracy of FSM

MEMS FSM is also susceptible to external force heat, electromagnetic radiation and other effects, whether quasi-static or resonant mode of operation, need closed-loop control to achieve high-precision beam pointing. The integrated capacitance angle sensor monitors the twisting angle of the micromirror in real time.

For the space laser communication terminal, the carrier beam is usually emitted at the beam divergence Angle near the diffraction limit, but under the long-distance space transmission of thousands to tens of thousands of kilometers, the loss is more than 100dB. For the space vehicle load platform, the weight, size and power consumption of the laser communication terminal are generally limited, and the carrier transmission power, optical receiving aperture and laser communication link margin are also limited. For the laser beam obeying the Gaussian distribution, the light intensity is the highest at the optic axis. If there is a certain error in the optic axis of the laser beam, The intensity of the laser beam at the receiving end will decrease in a Gaussian distribution.

In long-distance laser communication, the two terminal optical axes of communication need to be aligned as much as possible, and the execution accuracy of FPA, PAA and nutation mechanism used for fine tracking branches naturally requires higher accuracy. When the tracking error of the laser communication terminal is 3 μ rad, the optical link power loss is 1.22dB. When the tracking error is 6 μ rad, the optical link power loss is 4.89dB. The mechanical execution accuracy of traditional PZT and electromagnetic FSM can reach the order of sub-micro radians, and the link loss of laser communication due to alignment error is also acceptable in engineering, so the closed-loop control accuracy of MEMS FSM needs to reach the order of sub-micro radians.

For the selected MEMS FSM, the working angle resolution is also related to the angle deflection range, the measurement accuracy of the feedback sensor and the electronic drive processing circuit. Assuming that the MEMS FSM has a full range of 2mrad, the implementation accuracy of the mechanism is required to be 1.5 μ rad. If the self-closed loop of the mechanism adopts 16bit quantization, the digital control jump variable is required to be less than 50, which has engineering realizability.

For the resonant mode of MEMS FSM in the nutation coherent tracking system, the nutation half angle is generally in the order of 10 μ rad, and the coaxiality is adjusted in the order of 10 μ rad, so the execution accuracy is required less than or equal to 1 microradian. For quasi-static operation, execution accuracy is required less than or equal to 3 microradian.

3.5 Surface roughness of scanning mirror

The wavefront distortion caused by the transmitting optical unit is directly caused by the wavefront aberration of the fast mirror. The wavefront power distribution of the laser beam fluctuates or dithers due to the influence of the wavefront quality of many laser emitting units. The total wavefront error of the transmitting unit is the mean square sum of each component, and equation (4) is the approximate expression of the light intensity loss caused by distortion in the paraxial optical system

$$L_{wf} = e^{-(k\sigma)^2} \quad (4)$$

For most space laser communication systems, the wavefront error of the laser emitting unit is usually required to be less than $\lambda/10$, and the corresponding power loss is: $L_{wf} = e^{-[(2\pi/\lambda) \cdot (\lambda/10)]^2} = 0.67 = -1.74dB$.

The manufacturing process of MEMS pendulum mirror determines that the thickness of the mirror is very thin. In the process of high-speed swing, the optical quality of the reflected spot in the dynamic motion is not bad, and the static deformation and dynamic deformation parameters of the pendulum mirror are used to evaluate this performance[11-12]. Static deformation of MEMS scanning mirror in static state Δ_{sta} Mainly depends on the mirror curvature R and mirror size D.

$$\Delta_{sta} = R \cdot (1 - \cos(\arcsin(D \cdot R/2))) \quad (5)$$

It can be seen from equation (5) that under the requirement of specific static deformation, with the increase of mirror area, higher requirements are put forward for the curvature radius of the scanning mirror. The current manufacturing technology level can not solve the yield problem of large mirror mems, so it limits the frontal area of the scanning mirror to be larger.

In the dynamic state, according to the diffraction limit of the spot, the dynamic deformation Δ_{dyn} [12-14] is limited to less than the incident laser $\lambda/10$.

$$\Delta_{dyn} = 0.217 \cdot \rho f^2 D^5 \theta_{mesh} / Et^2 \quad (6)$$

Where ρ represents density, f represents vibration frequency, E is Young's modulus, t is mirror thickness, and θ_{mech} is the mechanical scanning angle of the scanning mirror. The size of the mirror area directly determines the static and dynamic variables. The large-area scanning mirror increases the mass and the inertia moment of the driver, which directly affects the dynamic deformation of the pendulum mirror.

For the application of space laser communication system, the mirror orientation is better than $\lambda/30@632nm$.

3.6 Reliability requirements for MEMS FSM

If the MEMS FSM is used for fine tracking mechanism and advance sight mechanism, the strong continuous laser amplified by the optical amplifier continues to pass through the reflecting mirror, and the continuous optical power is about 33~40dBm according to the different distance of the communication link. If the MEMS FSM is used for nutation coherent tracking, the laser terminal receives the weak signal light that is lost in a long distance space through the reflecting mirror, and the continuous power is about -40~-60dBm.

The optical coatings of different materials on the surface of the MEMS FSM limit the optical reflectivity and the peak laser power that can be withstand. The high laser energy will cause irreversible damage to the optical coating. The MEMS FSM used in the fine tracking and advance sight mechanism cannot implement thermal control cooling due to its small size. The special process of MEMS FSM determines its sensitivity to heat source, and the absorption of continuous laser energy by optical coating in space environment causes thermal deformation or even burnout of MEMS FSM, which directly affects the beam pointing accuracy. Common optical coatings are pure aluminum film and pure gold film, respectively, with more than 90% reflectivity in the near infrared band, which can withstand the highest peak power of 10kW pulsed laser up to about an order of magnitude. Assuming that the surface of the MEMS FSM is gold-plated and the near-infrared optical reflectivity is 99%, the heat absorbed by the scanning mirror after the 37dBm light beam is reflected by the scanning mirror is 0.05W, and the heat is huge after long-term work accumulation. Taking into account the space vacuum environment and the structure of the MEMS scanning mirror, the reflectivity of the scanning mirror is preferably 99%, and a good heat dissipation design is carried out for the scanning mirror to reduce the heat dissipation burden of the MEMS FSM.

Based on the above analysis, the technical parameters of the MEMS FSM used by the space laser communication terminal are shown in Table 3-1.

Table 3-1. MEMS FSM technical requirements for space laser communication applications

Name	Linear mode	Resonant mode
Bandwidth(closed loop)	$\geq 1kHz$	$\geq 10kHz$
Size of scanning mirror	$\geq \Phi 5mm$	$\geq \Phi 5mm$
Surface roughness of scanning mirror	$\lambda/30@632nm$	$\lambda/30@632nm$
Optical reflectivity	99%@1550nm	99%@1550nm
Laser damage threshold	$\geq 3W$	$\geq 1mW$
Power dissipation	$< 200mW$	$< 200mW$
Accuracy of beam scanning	$< 10\mu rad$	$< 3\mu rad@ \geq 100\mu rad$
Scanning angle	$\geq \pm 10mrad$	$\geq \pm 1mrad$
Temperature	15~25℃	15~25℃

4. CONCLUSION

Two-dimensional MEMS scanning mirrors for applications in space laser communication systems: The mechanical scanning angle needs to be greater 10mrad, the scanning frequency can reach ten kilohertz, the execution accuracy is of the order of micro radian, the mirror size is 5mm, the curvature radius is meter, the static mirror flatness is better than

30nm,the dynamic deformation is less than 50nm,the MEMS scanning mirror adopts metal reflecting film,and the laser power can be withstand more than watts.

REFERENCES

- [1] ISHOLAF,CHO M. Experimental Study on Photodiode Array Sensor Aided MEMS Fine Steering Mirror Control for Laser Communication Platforms[J]. IEEE Access, 2021, 9:100197-100207.
- [2] Yang C.W., Chen M.,Liu X.N., et al. Current status and development trends of minisatellite laser communication terminal technology[J], Journal of Telemetry, Tracking and Command, 2021.42(3):1-7.
- [3] Xia F.Y., Chen X., Chen A.H.,et al. Spaceborne Milliaturized Laser Commulication Terminal Current Situation and Development Trend[J].SPACE ELETRONIC TECHNOLOGY,2020,3:73~80.
- [4] Zhu W.H., Wang Y., Wang L.H., et al. Research progress of reliability of MEMS fast steering mirror for satellite laser communication[J], 2023.52(9): 20230179-1-13.
- [5] Yu Z.H., Wang L.H., Wang Y., et al. Research progress of fast steering mirror for satellite laser Communication[J/OL]. Laser & Optoelectronics Progress.
- [6] TAYLOR M., ROYCHOWDHURY A., MALDONADOS, et al. Polar Orbiting Infrared. Tracking Receiver (POINTR) [M]. 2019:13 pp.-13 pp.
- [7] CIERNY O.,CAHOYKL.On-orbit beam pointing calibration for nanosatellite laser communications[J].Optical Engineering. 2019.58(4).
- [8] Miner A., Milanovic V. High temperature operation of gimballess two axis micromirrors[C]//2007 IEEE/LEOS International Conference on Optical MEMS and Nanophotonics. IEEE, 2007:91-92.
- [9] Willstatter L., Mahon R., Ghiorzi J., et al. Characterization of fast-steering mirrors at both high and low temperatures [C]//Proceedings of SPIE, 2020, 11272: 112721K.
- [10]Hu H., Liu L., Li C., et al. Research on composite scanning strategy for high-probability acquisition of deep space cooperative targets[J].Chinese Space Science and Technology, 2022,42(3):58-66.
- [11]Li F.H., Zhao J.G., Du X.P., et al. Research status and trend analysis of beam deflection technology based on space laser communication[J].Infrared and Laser Engineering, 2023, 52(10):20230004-1-23.
- [12]Zhang X.L., Wang W., Wu J.B., Research on Fast Steering Mirror Control Technology of Laser Communication System[J]. Semiconductor Optoelectronics, 2021, 42(02): 289-294.
- [13]Wang Z., Cheng X.M., Research progress and development trend of fast steering mirror[J]. Journal of Applied Optics, 2019, 40(03): 373-379.
- [14]Zhao B.Q., Yu X.N., Dong Y., et al. Tracking Performance of Mirrors in Space Laser Communication Networking[J]. Laser & Optoelectronics Progress, 2021, 58(9):0906007.